Non-Semaev decompositions for prime-field ECDLP

Non-Semaev decompositions for prime-field ECDLP: elliptic nets, division polynomials, isogeny relations

Date. 2026-11-13. **Type.** Search round and analysis (no experiment run). **Standing.** Every new statement here is an unverified assertion, **[pcc:unverified]**: no pcc claim is filed. Quoted results keep the standing their source files give them, which is mostly `[pcc:unverified]`, toy-scale and single-run.

Question

The user's question was whether a decomposition for `E/F_p` that does not use Semaev summation polynomials could work. In it, some algebraic structure (elliptic nets, division-polynomial structure, or isogeny-based relations) would take the place of the subfield that Weil descent uses over `F_{q^n}`. The user called it speculative, with no supporting evidence yet.

Result

1. **All three suggested structures have already been tried in this lab and closed at the exponent level.** The map (§2) gives the paths and the failure mechanisms.
   - Elliptic nets reduce to the rank-1 EDS, which Lauter–Stange show is equivalent to ECDLP.
   - Division polynomials are closed by an exact obstruction: prime order makes `E(F_p)[m] = {O}`, and `[m]` is an automorphism.
   - Isogenies are closed for three reasons:
     - the isogeny-fibre programme's Thm 15.1 says they add nothing in the GGM;
     - the pullback of `S_3` is a product of kernel-twisted `S_3`;
     - the PKM isogeny tower costs 1.00–1.33× a plain base of equal size.
2. **A non-Semaev decomposition has two independent parts** (§3):
   - a *relation source*: an encoding of addition, or an algebraic correspondence;
   - a way to *attach the target*: membership of the target's parts in the factor base.

   The barrier map (`research/notes/ecdlp-general/RESEARCH_PRIME_FIELD_BARRIER_MAP_20261104.md`) already shows that changing the addition encoding does not move the cost, and that membership is blocked (L1–L3, archimedean, p-adic, height).
3. **New here: Lemma T, the target-connection bound** (§4). It covers every relation source that avoids membership:
   - Hecke correspondences;
   - Heegner/Kolyvagin norm relations;
   - isogeny-graph identities;
   - elliptic-net identities.

   All of these give relations *among factor-base points*, so at best a factor base with known logs. Attaching a target generically then costs at least `n/(2B)` group operations, so the total is at least `c·B + n/(2B) ≥ √(2cn)`.
   - **Exponent.** It stays at `1/2` whatever the relations cost.

    - **Constant.** Negation-rho (≈ `0.886√n`) is beaten only if a known-log base point costs `c < 0.39` group operations, including its share of the linear algebra.
    - **Linear algebra.** Unstructured linear algebra alone pushes the cost to `n^{2/3}`.

  This closes the one live thread in the "isogeny-based relations" direction (Hecke/CM class points, recorded as "open, not promising, unpriced") at the exponent level, before any pricing.
4. **What survives** (§5). Only one proposal class passes every filter: an algebraic solver whose working degree stays bounded on a membership tower. The elliptic-net idea's surviving form is the Riemann–Roch (Frobenius–Stickelberger) addition encoding on such a tower.
    - It was stage-level positive at `m = 3`.
    - It is undecided at `m = 5`.
    - It is a constant-level lever unless the tower degree flattens.

  §6 gives the cheapest discriminating next step.

1. Sources checked

- **research-kb.** Not available in this session: the MCP server was not connected, and the `scripts/research-kb` CLI needs a `crypto-autoresearcher` virtualenv that is not on this host. No `source_id` can be recorded. This search used the files below.
- **Lead atlas.** `crypto-autoresearcher` was cloned read-only and sparsely, at `knowledge/leads/ecdlp/`. Every record there is vault-encrypted (`VLT1` header) and unreadable here. Its content was taken from the plaintext copies `crypto:research/prime_field_lead_assessment_20261107/atlas_survey.md` and `crypto:research/lead_sweep_20261110/*/RESULT.md`.
- **Read in full or in the relevant sections:**
    - `crypto:research/notes/ecdlp-general/RESEARCH_PRIME_FIELD_BARRIER_MAP_20261104.md` (§§0–2, 5, 6)
    - `crypto:research/semaev_alternatives_audit_20261104/REPORT.md` (§3 gate, §9 disposition)
    - `crypto:research/notes/index-calculus/RESEARCH_PKM_TOWER_ORACLE.md` (status block)
    - `crypto:src/cryptanalysis/f4_fp_tower.rs` (header: Kummer, Dickson and 2-isogeny tower rules)
    - `crypto:research/lead_sweep_20261110/LEAD-EXO-lattes-xline-dynamics/RESULT.md`
    - `cryptanalysis:experiments/hecke-sparse-target-20261030/{RESULT,PAIR_REUSE_RESULT}.md`
    - `crypto:research/prime_field_lead_assessment_20261107/crypto_survey.md`
- **Sweep.** A read-only sweep of both repositories by a delegated agent found the files in §2. Its file:line citations are reproduced there. The paths were checked for existence; the quoted numbers were not re-derived.

2. Prior work on the three suggested structures (and neighbours)

structure	path	recorded verdict
EDS / elliptic net, residue χ, 2-D net	`crypto:research/notes/cm-isogeny/RESEARCH_EDS_RESIDUE.md` (:278, :391, :430, :726)	falsified for sub-√m
net / Somos sieve, short annihilator	`crypto:research/lead_sweep_20261110/LEAD-EXO-elliptic-net-eds/RESULT.md:11`; P1540 audit in `research/torsion_division_derivative_20261031/PROTOCOL.md`	falsified at toy scale
ψ_m torsion, level-set and	`crypto:research/pair46_factor_bases_20261015/07-torsion-division/RESULTS.md:5`	exact obstruction

structure	path	recorded verdict
preimage bases		
division-derivative parity oracle (H109)	`crypto:research/torsion_division_derivative_20261031/`	falsified for the frozen family (0/28 exact formulas)
FGHR 2-/4-torsion symmetrisation	`crypto:research/prime_s5_small_root_leads_20261028/README.md:545`	closed
isogeny fibres, pullback, transport	`crypto:RESEARCH_ISOGENY_FIBERS_ECDLP.md` (Thm 4.2; Thm 15.1 at :1472)	constants only
isogeny-class yield, conductor gaps	`crypto:research/isogeny_class_ic_yield_20261102/`; `drastic_prime_conductor_gap_20261012/`	null
PKM 2-isogeny (Vélu) tower base	`crypto:research/pkm_isogeny_tower_ecbench_20261015/`; `RESEARCH_PKM_TOWER_ORACLE.md`	falsified as a gain
Vélu-invariant / Q-curve bases	`crypto:research/notes/index-calculus/RESEARCH_GLV_INVARIANT_FACTOR_BASES.md:195`	falsified
isogeny + genus-2 cover transfer	`crypto:research/chained_prime_transfer_20261111/RESULTS.md:36`	falsified
Hecke / CM class-point relations	`cryptanalysis:experiments/hecke-sparse-target-20261030/PAIR_REUSE_RESULT.md`	open, not promising, unpriced
Nagao / Riemann–Roch on `F_p`	`crypto:research/split_prime_ecdlp_20261107/followups/p1_rr_dickson/RESULTS.md`; `research/prime_norm_rank_20261112/REPORT.md`	stage-positive at `m=3`, open at `m=5`
xedni / global lift / p-adic	`crypto:research/jksst_xedni_audit_20261101/`; `lead_sweep/LEAD-TRF-kernel-of-reduction-lemma`	closed (conditional end-to-end)
Miller / pairing cocycle	`crypto:research/lead_sweep_20261110/LEAD-EXO-miller-pairing-cocycle/RESULT.md:12`	falsified at toy scale

3. Anatomy of any decomposition: relation source × target attachment

Over `F_{q^n}` the subfield does two jobs. Over `F_p` one has to ask which of the two a candidate structure replaces.

- **(a) A small set with a short presentation.** `F_q` has `q = p^{1/n}`-ish elements, and Frobenius makes `x^q = x` linear after descent. This gives membership with polylogarithmic presentation cost and *conjugate equations* (barrier map L2).
- **(b) Relations that involve the target.** The descended system contains `x_R`, so every solution is a decomposition of `R` (or of `aG+bR`).

Relation counts are not what the subfield supplies. Mean yield is `C(B+m−1,m)/N` for *every* base (KN-FIND-007, 288/288 cells), so subfield bases are random-like in their counts. The subfield's contribution is that solutions can be *found*.

How the user's structures map onto (a) and (b):

structure	what it can replace	blocked by
elliptic-net polynomial `Ψ_{(1,…,1)}` = Frobenius–Stickelberger / Riemann–Roch determinant	the *addition encoding* inside (a)+(b)	the encoding is not the cost (barrier map §1: group-law `d_reg = 2` everywhere; Edwards and shape-matched nulls equal); membership L1–L3
division-polynomial towers (x-maps of `[2]` on an auxiliary `E'`, Lattès maps)	the *presentation* in (a)	already the PKM family; L3 ("no fourth presentation"); tower `D` slope
`[m]`, ψ_m on `E` itself	(a)	exact obstruction (prime order)
isogeny transport `E → E'`	relabelling only	Thm 4.2 / Thm 15.1
Hecke / Heegner / Kolyvagin / isogeny-graph identities	a *relation source with no membership at all*	**Lemma T** (they do not supply (b))

4. Lemma T (target connection) [pcc:unverified, proof sketch]

Setting. `⟨G⟩` is cyclic of prime order `n`. Some procedure, generic or not, outputs `B` group elements `Y_1, …, Y_B` with their logarithms `s_i = log_G Y_i`. Its total cost is `C_base`, measured in group-operation equivalents and including any linear algebra that turns relations into logs. The target `R` is then used only through group operations: additions, comparisons of encodings, and so on.

Claim. Any such algorithm that outputs `log_G R` with constant probability makes `q = Ω(min(√n, n/B))` group operations involving `R`. Hence the total cost satisfies `C_base + q ≥ c·B + n/(2B) ≥ √(2cn)`, where `c = C_base/B`; the `n/(2B)` uses the negation map.

Sketch. 1. In Shoup's generic group model, elements derived from `R` have the form `a·R + b·G` with known `(a, b)`. 2. A logarithm is learned only from a collision between two encodings whose formal labels differ. Either both involve `R`, or one involves `R` and the other is a known-log element (the `B` inputs and anything derived from them). 3. With `q` operations involving `R`, the probability of any such collision is at most `q(q + B + q')/n`, where `q'` counts the operations that do not involve `R`. 4. Constant success therefore needs `q ≳ n/(B + q)`, which gives `q ≳ min(√n, n/B)`. 5. Known-log inputs are exactly baby steps that were not paid for in group operations, so the bound is BSGS with free baby steps. 6. The negation map halves the giant-step count. Its usual form folds ±.

Consequences. - **Exponent.** For every relation source that produces relations among base points only, the exponent stays `1/2`. This includes Hecke correspondences on `X_0(N)`, Heegner points with Kolyvagin norm relations `Tr_{cℓ/c} y_{cℓ} = a_ℓ y_c`, isogeny-graph adjacency identities, and elliptic-net recurrences. - **Constant.** `min_B (cB + n/(2B)) = √(2cn)`, against negation-rho `√(πn/4) ≈ 0.886√n`. A gain needs `c < π/8 ≈ 0.39`. One known-log base point must then cost less than 0.39 group additions, including its share of the linear algebra, with `B ≈ √n` points stored. - **Linear algebra.** Generic sparse linear algebra on `B` unknowns costs `≳ B²`, which forces `B ≲ n^{1/3}` and a total `≳ n^{2/3}`, worse than rho. Linear algebra in `O(B log B)` needs structured relations. Heegner points of a fixed discriminant are a torsor under `Cl(D)`. Hecke relations are then a Cayley-graph convolution, diagonalised by class-group characters. The log function is supported on the characters with `χ(ι) + χ(ι)^{-1} ≡ a_ℓ (mod n)` for every split `ℓ` used. That explains the few free logs left in the Hecke records (9–12, 45). Each free log is itself a DLP instance, which is the opposite of help. - **The only escape is non-generic use of `R`.** That means using `R`'s coordinates to decompose it over the base, which is membership, and L1–L3 (barrier map §2) then apply. Multi-target amortisation stays inside the generic preprocessing bounds (Bernstein–Lange; Corrigan-Gibbs–Kogan).

Not covered. - **Coordinate-revealing oracles.** These are not GGM-simulable, and their atlas lead is open (`LEAD-EXO-coordinate-revealing-oracles`). - **Free relations that already involve `R`.** No construction of such relations is known, because `R` is arbitrary. - **Rigour.** The sketch is the standard Shoup collision

argument with extra known-log elements. It has not been formalised.

5. What survives

The barrier map's filter (§5) rejects at proposal time any prime-field idea that does not name which of L1, L2/L3, k-SUM, p-adic splitting and height it evades. Lemma T adds a sixth item: **(T) relations among base points only**. Every user-suggested structure is caught by at least one item (§3 table). Three classes are not caught.

1. **Bounded working degree on a membership tower.** Is there an algebraic solver whose degree `D` stays bounded (at 256 bits, `D ≲ 10` at `N ≈ 256` under the quadratic cost model)?
   - Measured slope: 0.125–0.15 per tower level.
   - The k-SUM heuristic predicts `D ≳ 0.04·N + c`.
   - The decisive question is whether `D` flattens.
2. **The Riemann–Roch / net encoding on such a tower.** This is the elliptic-net direction's surviving form.
   - At `m = 3` it beat the chain system and traded against flat Semaev on width, as recorded in `p1_rr_dickson/RESULTS.md` [pcc:unverified].
   - At `m = 5, t = 4` it is undecided: Spot interruptions killed every attempt, and the chain arm hit a 56 GiB cap at `D ≥ 9`.
   - The barrier map predicts it cannot move the exponent: the encoding is not the cost.
3. **Coordinate-revealing (non-generic) oracles.** Open; no construction is on file.

6. Recommended next step (not run)

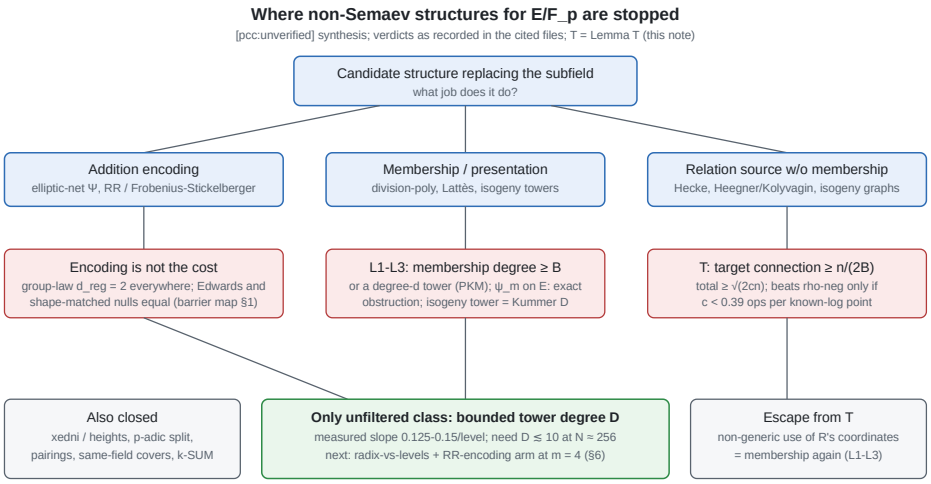
Given the user's frame, the most discriminating cheap test is the barrier map's §6.1 **radix-vs-levels** discriminator, extended with an encoding arm.

- **Question.** At equal `|V|`, is the tower `D` governed by the number of levels or by `log₂|V|`? Does the Riemann–Roch encoding lower `D` relative to `S_{m+1}` on the same tower?
- **Build.**
  - `f4_fp_tower` supports only degree-2 steps (`SquareRule`; square-free `u128` masks).
  - A radix-3 arm needs a `CubeRule`, 2-bit exponents, and degree as an exponent sum (barrier map §6.1, checked against the code there).
  - The Riemann–Roch arm can reuse the `p1_rr_dickson` system builder.
- **Cells.** `m = 4` (the first arity where the gate `s < m − 3` can matter), `|V| ∈ {2^6, 2^12}`, 3 primes × 2 planted targets, with a shape-matched null per arm.
- **Pre-registered prediction (this note).** `D_RR = D_Semaev` and `D_radix3 ≥ D_radix2 + 1` at equal `|V|`.
- **Falsifiers.**
  - `D_RR ≤ D_Semaev − 1` at `|V| = 2^12` with matrix width not larger; or
  - `D_radix3 = D_radix2` with multiply-adds lower by ≥ 2×.

  Either result would be the first evidence that the presentation or the encoding moves the cost.
- **Route.** Per CLAUDE.md, x86 runs go to isolab-eks with a frozen task card, a resume checkpoint and a `bench.record`. The `m = 5` Riemann–Roch cell needs a non-preemptible or CRIU-resumable host.

7. Limits

- No new measurement was made.
- Lemma T is a proof sketch and is not formalised.
- The prior-work verdicts are those recorded in the cited files, mostly toy-scale and `[pcc:unverified]`.
- The lead atlas could not be read directly (it is encrypted here), and research-kb was unavailable. A later session with both should repeat the "not tried" checks in §2 before citing them.



Open, not covered by T: coordinate-revealing (non-GGM) oracles; relations that already involve R (no construction known).

Sources: RESEARCH_PRIME_FIELD_BARRIER_MAP_20261104.md; semaev_alternatives_audit_20261104; RESEARCH_PKM_TOWER_ORACLE.md; RESEARCH_EDS_RESIDUE.md; pair46 07-torsion-division; RESEARCH_ISOGENY_FIBERS_ECDLP.md; hecke-sparse-target-20261030.

Barrier tree